PREPARING A PUSHOUT SQUARE BY MATCHING AFFINE OPENS ALONG THE CLOSED PIECE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 19726-19755, immediately following the statement at 19719-19724.

Editorial note. Two cross-chapter pointers (Limits, Morphisms) plus the chapter's own situation are recorded in assembly.external_references.

Editorial note. The author's phrase 'Now we (in some sense) repeat the argument' is retained verbatim.

Allowed standard prerequisites (editorial).

- The pushout situation of the section. chapter situation situation-pushout-along-closed-immersion-and-integral, cited in the statement. Fixes the closed immersion i : Z -> X and integral j : Z -> Y whose pushout is being prepared.
- Integral morphisms are universally closed. Morphisms, lemma morphisms-lemma-integral-universally-closed, cited at proof line 19731. Makes j : Z -> Y closed, which is what lets an affine neighbourhood of y inside i^{-1}(U) be found.
- Ample profinite sets in a principal affine open. Limits, lemma limits-lemma-ample-profinite-set-in-principal-affine, cited in the proof. Supplies the principal open D(g) inside the previous affine chart used in the second pass of the argument.

Source reference. Original reference 0E21: focused-seeds/geometry/S9-morph-prepare-pushout-closed-integral/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01WM: focused-seeds/geometry/S9-morph-prepare-pushout-closed-integral/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ECI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19708–19708. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0ECI then for $y \in Y$ there exist affine opens $U \subset X$ and $V \subset Y$ with $i^{-1}(U) = j^{-1}(V)$ and $y \in V$.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $y \in Y$. Choose an affine open $U \subset X$ such that $j^{-1}(\{y\}) \subset i^{-1}(U)$ (possible by assumption). Choose an affine open $V \subset Y$ neighbourhood of y such that $j^{-1}(V) \subset i^{-1}(U)$. This is possible because $j : Z \to Y$ is a closed morphism (Morphisms, Lemma 01WM) and $i^{-1}(U)$ contains the fibre over y. Since j is integral, the scheme theoretic fibre Z_y is the spectrum of an algebra integral over a field. By Limits, Lemma 0E21 we can find an $\overline{f} \in \Gamma(i^{-1}(U), \mathcal{O}_{i^{-1}(U)})$ such that $Z_y \subset D(\overline{f}) \subset j^{-1}(V)$. Since $i|_{i^{-1}(U)} : i^{-1}(U) \to U$ is a closed immersion of affines, we can choose an $f \in \Gamma(U, \mathcal{O}_U)$ whose restriction to $i^{-1}(U)$ is $\overline{f}$. After replacing U by the principal open $D(f) \subset U$ we find affine opens $y \in V \subset Y$ and $U \subset X$ with

$$j^{-1}(\{y\}) \subset i^{-1}(U) \subset j^{-1}(V)$$

Now we (in some sense) repeat the argument. Namely, we choose $g \in \Gamma(V, \mathcal{O}_V)$ such that $y \in D(g)$ and $j^{-1}(D(g)) \subset i^{-1}(U)$ (possible by the same argument as above). Then we can pick $f \in \Gamma(U, \mathcal{O}_U)$ whose restriction to $i^{-1}(U)$ is the pullback of g by $i^{-1}(U) \to V$ (again possible by the same reason as above). Then we finally have affine opens $y \in V' = D(g) \subset V \subset Y$ and $U' = D(f) \subset U \subset X$ with $j^{-1}(V') = i^{-1}(V')$. $\square$